

# Amines

NCERT Class XII Chemistry · Chapter 9 · Amines

## Amines — Complete Notes

6 marks in the board paper (CBSE 2026-27 curriculum), and a chapter that is mostly reactions you can predict once you know one fact: the nitrogen has a lone pair of electrons. That lone pair makes amines bases, makes them nucleophiles, and decides what each reagent does to them. Every new word is tagged [exposure] and explained from scratch. The two big payoffs are the basicity order and the diazonium-salt conversions — they carry most of the marks.

### What amines are, and where they matter — *ammonia with its hydrogens swapped for carbon groups*

**[exposure]** An **amine** is an organic compound you get by replacing one or more hydrogen atoms of **ammonia**  $\text{NH}_3$  with alkyl or aryl groups. (An **alkyl** group is a saturated carbon chain such as  $\text{CH}_3-$ ; an **aryl** group is a benzene-ring group such as  $\text{C}_6\text{H}_5-$ .) Amines matter because the same  $-\text{NH}_2$ -type unit sits inside proteins, vitamins, hormones and many drugs. Example: methylamine  $\text{CH}_3\text{NH}_2$  is ammonia with one hydrogen replaced by  $\text{CH}_3$ .

Where they turn up, and what the examiner can ask:

- Biologically active amines: **adrenaline** and **ephedrine** contain a secondary amino group and raise blood pressure.
- **Novocain** — a synthetic amine used as a local anaesthetic in dentistry.
- **Benadryl** — an antihistamine (it treats allergic reactions) with a tertiary amino group.
- **Quaternary ammonium salts** are used as **surfactants** (they lower the surface tension between two liquids that do not mix).
- **Diazonium salts** are intermediates for making a huge range of organic compounds, **dyes** included.

**Examiner asks:** nothing hard from this section, but the drug names are one-mark recall.

### Classification and structure — *count the carbon groups on nitrogen; then why the shape is a pyramid*

**[exposure]** Amines are sorted by **how many hydrogens of ammonia were replaced**. This count decides almost every reaction later.

| Type                    | Replaced | General formula       | Example                    |
|-------------------------|----------|-----------------------|----------------------------|
| Primary ( $1^\circ$ )   | one H    | $\text{RNH}_2$        | $\text{CH}_3\text{NH}_2$   |
| Secondary ( $2^\circ$ ) | two H    | $\text{R}_2\text{NH}$ | $(\text{CH}_3)_2\text{NH}$ |
| Tertiary ( $3^\circ$ )  | three H  | $\text{R}_3\text{N}$  | $(\text{CH}_3)_3\text{N}$  |

**[exposure]** A **quaternary ammonium salt** has *four* carbon groups on nitrogen, replacing all four hydrogens of the ammonium ion  $\text{NH}_4^+$ :  $\text{R}_4\text{N}^+\text{X}^-$ . The nitrogen carries a positive charge and has no lone pair left to give away, which is why such salts are not bases.

**[exposure]** **Simple versus mixed.** A secondary or tertiary amine is **simple** if all its carbon groups are the same (diethylamine, trimethylamine, diphenylamine) and **mixed** if they differ (ethylmethylaniline, ethylphenylamine).

### Structure — why the shape is a pyramid

The nitrogen of an amine is  $sp^3$  **hybridised**: one *s* and three *p* orbitals blend into four equal hybrid orbitals. Three of them make sigma bonds (to H or R); the fourth holds the **lone pair**. A lone pair repels the bonded pairs more than they repel each other, so the shape is a **pyramid** (the same as ammonia) and the bond angle is squeezed below the tetrahedral  $109.5^\circ$ . In trimethylamine it is about  **$108^\circ$** .

**Examiner asks:** classify given amines as primary, secondary or tertiary — count the groups on nitrogen (NCERT intext 9.1).

## Nomenclature — *common names for the pattern; IUPAC "alkanamine" for the marks*

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### Common names

Name the alkyl group(s), then write **amine**. Same groups repeated: **di-**, **tri-**. Different groups:

**alphabetical order.**  $\text{CH}_3\text{NH}_2$  = methylamine;  $(\text{CH}_3)_2\text{NH}$  = dimethylamine;  $(\text{CH}_3)_3\text{N}$  = trimethylamine;  $\text{C}_2\text{H}_5\text{NHCH}_3$  = ethylmethylaniline.

### IUPAC names

1. **Primary:** drop the final "e" of the alkane and add **amine**:  $\text{CH}_3\text{NH}_2$  = **methanamine**;  $\text{CH}_3\text{CH}_2\text{NH}_2$  = **ethanamine**;  $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$  = **propan-2-amine** (number the carbon that carries  $-\text{NH}_2$ ).
2. **Secondary and tertiary:** the **longest chain** gives the parent name. The smaller group(s) on nitrogen become substituents written with the locant N:  $\text{CH}_3\text{NHC}_2\text{H}_5$  = **N-methylethanamine**.
3. Two identical small groups: **N,N-** prefix.  $(\text{CH}_3)_2\text{NC}_2\text{H}_5$  = **N,N-dimethylethanamine**. Different small groups: alphabetical, each with its own N: **N-ethyl-N-methyl...**
4. **Two  $-\text{NH}_2$  groups:** keep the "e" and add **diamine**:  $\text{H}_2\text{N}-\text{CH}_2\text{CH}_2-\text{NH}_2$  = **ethane-1,2-diamine**.

### Aromatic amines

$\text{C}_6\text{H}_5\text{NH}_2$  is **aniline** (IUPAC also accepts **benzenamine**). A methyl on the ring gives **toluidine** (ortho, meta, para). With IUPAC numbering the  $-\text{NH}_2$  carbon is 1: so 2-bromoaniline, and 2,4,6-tribromoaniline. Groups on nitrogen use N:  $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$  = **N,N-dimethylaniline**.

**Trap:** in a secondary or tertiary amine the *longest* chain is the parent and every other carbon group is an **N-** substituent. The smaller group goes in front with N, not the bigger one. A tertiary amine with two ethyls and one methyl on N is **N-ethyl-N-methylethanamine**, not "triethyl...".

**Examiner asks:** write the IUPAC name of a given amine — a standing 1-marker (2017, 2016, 2022) and a 2-marker (2023). See the PYQ section.

## Isomers of an amine formula — how C<sub>4</sub>H<sub>11</sub>N gives eight isomers, and the four kinds of isomerism they show

Draw isomers by first fixing the *type* of amine, then moving the groups. For C<sub>4</sub>H<sub>11</sub>N (NCERT intext 9.2) there are **eight**:

| Type          | Isomers (IUPAC)                                                                 |
|---------------|---------------------------------------------------------------------------------|
| Primary (4)   | butan-1-amine · butan-2-amine · 2-methylpropan-1-amine · 2-methylpropan-2-amine |
| Secondary (3) | N-methylpropan-1-amine · N-ethylethanamine · N-methylpropan-2-amine             |
| Tertiary (1)  | N,N-dimethylethanamine                                                          |

**[exposure] Isomers** are different compounds with the same molecular formula. Amines show four kinds:

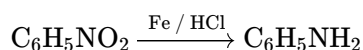
- **Chain isomerism** — the carbon chain differs; the –NH<sub>2</sub> position is the same (butan-1-amine and 2-methylpropan-1-amine).
- **Position isomerism** — the position of the functional group differs (butan-1-amine and butan-2-amine).
- **Metamerism** — same functional group, but the alkyl groups on either side of it differ. It happens in secondary amines (N-methylpropan-1-amine, N-ethylethanamine, N-methylpropan-2-amine).
- **Functional isomerism** — the amine class itself differs: primary, secondary and tertiary amines of one formula are functional isomers of each other.

**Examiner asks:** C<sub>3</sub>H<sub>9</sub>N — four isomers: propan-1-amine, propan-2-amine, N-methylethanamine, N,N-dimethylmethanamine. Which of them release nitrogen with nitrous acid? Only the *primary* ones (NCERT intext 9.8); see the nitrous-acid section.

## Preparing amines — six routes — and a one-line reason to prefer each

### 1 · Reduction of nitro compounds

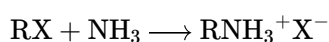
**[exposure]** A **nitro compound** carries the group –NO<sub>2</sub>. Reducing it (adding hydrogen, removing oxygen) gives the amine. Two ways: **catalytic hydrogenation** (hydrogen gas with a finely divided metal catalyst — Ni, Pd or Pt) or **a metal in acid** (Sn/HCl, Fe/HCl).

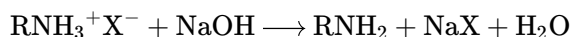


**Why Fe + HCl is preferred:** the FeCl<sub>2</sub> it makes hydrolyses and gives back HCl, so only a *small* amount of acid is needed to start the reaction and it keeps itself going.

### 2 · Ammonolysis of alkyl halides

**[exposure]** **Ammonolysis** means breaking a bond using ammonia. When an alkyl (or benzyl) halide is heated with ethanolic ammonia in a sealed tube at 373 K, the lone pair on NH<sub>3</sub> attacks the carbon and kicks out the halide: nucleophilic substitution. The product is a substituted ammonium salt; a base (NaOH) then frees the amine.

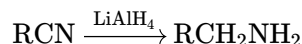




**The drawback:** the new amine is itself a nucleophile, so it attacks another RX, then another — you get a **mixture of 1°, 2°, 3° amines and a quaternary salt**. To make mainly the *primary* amine, use a **large excess of NH<sub>3</sub>**. Reactivity of the halide: RI > RBr > RCl.

### 3 • Reduction of nitriles

**[exposure]** A **nitrile** is R–C≡N (a cyanide group on carbon). Reducing it with LiAlH<sub>4</sub>, sodium–mercury alloy in ethanol, or H<sub>2</sub>/Ni gives a primary amine **with one carbon more** than the halide you started from — it is the route for **climbing up** an amine series.



### 4 • Reduction of amides

An **amide** is RCONH<sub>2</sub>. LiAlH<sub>4</sub> reduces it to the primary amine RCH<sub>2</sub>NH<sub>2</sub> (same carbon count).



### 5 • Gabriel phthalimide synthesis

**[exposure]** **Phthalimide** is a ring imide whose N–H is acidic. The **Gabriel synthesis** uses it as a clean source of a single –NH<sub>2</sub> group, so you avoid the mixture problem of ammonolysis. Steps: (1) phthalimide + ethanolic KOH gives the **potassium salt** (nitrogen now carries a negative charge); (2) the salt is heated with an **alkyl halide** — the nitrogen attacks the carbon, giving N-alkylphthalimide; (3) **alkaline hydrolysis** with aqueous NaOH releases RNH<sub>2</sub> and sodium phthalate. It gives **primary aliphatic** amines only.

**Trap:** Gabriel **cannot make aromatic primary amines** (aniline). An aryl halide does not undergo nucleophilic substitution with the phthalimide anion, because resonance gives the C–X bond partial double-bond character and it will not break. This is a standing 1-mark "give reason" (2017, 2020, 2025).

### 6 • Hofmann bromamide degradation

**[exposure]** **Hofmann bromamide degradation** converts an amide into a primary amine that has **one carbon less**. The amide is treated with Br<sub>2</sub> and aqueous or ethanolic NaOH; the R group migrates from the carbonyl carbon to nitrogen, and the carbonyl carbon is lost as sodium carbonate. It is the route for **stepping down** an amine series.



Example: CH<sub>3</sub>CONH<sub>2</sub> (ethanamide, 2 C) gives CH<sub>3</sub>NH<sub>2</sub> (methanamine, 1 C).

**Examiner asks:** the Hofmann equation (2019) and "amide that gives propanamine by Hofmann" (butanamide — NCERT Example 9.3). Nitrile reduction (2015). The Gabriel equation. All are 1- to 3-mark questions.

## Conversions: build and trim carbon chains — *three moves cover almost every "convert A to B" question*

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Before you pick reagents, **count the carbons** in the start and the product.

- **Same number of carbons:** reduce the nitro compound or amide, or use the Gabriel route.
- **One more carbon in the product:** go via the nitrile.  $R-X \xrightarrow{KCN} R-CN \xrightarrow{LiAlH_4} R-CH_2-NH_2$ .
- **One fewer carbon in the product:** go via the amide and Hofmann.  
 $R-COOH \xrightarrow{NH_3, \Delta} R-CONH_2 \xrightarrow{Br_2 / NaOH} R-NH_2$ .

Worked from NCERT (Examples 9.2, 9.3, 9.5 and intext 9.3):

Worked example ·  $CH_3CH_2Cl$  to  $CH_3CH_2CH_2NH_2$

Two carbons → three carbons, so go up through the nitrile:

$CH_3CH_2Cl \xrightarrow{KCN} CH_3CH_2CN \xrightarrow{LiAlH_4} CH_3CH_2CH_2NH_2$ . The cyanide ion replaces chlorine (nucleophilic substitution) and then  $LiAlH_4$  reduces the  $-CN$  to  $-CH_2NH_2$ .

Worked example · benzene to N,N-dimethylaniline

$C_6H_6 \xrightarrow{HNO_3 / H_2SO_4} C_6H_5NO_2 \xrightarrow{Fe / HCl} C_6H_5NH_2 \xrightarrow{2 CH_3Br} C_6H_5N(CH_3)_2$ . Nitrate, reduce, then alkylate twice; each methyl replaces one N-H.

Worked example ·  $Cl-(CH_2)_4-Cl$  to hexane-1,6-diamine

Four carbons → six, so go up through nitriles on both ends:

$Cl(CH_2)_4Cl \xrightarrow{2 NaCN} NC(CH_2)_4CN \xrightarrow{LiAlH_4} H_2NCH_2(CH_2)_4CH_2NH_2$ .

## Physical properties — *state, solubility, boiling point* — *three tables of "why"*

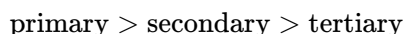
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- **State and smell.** Lower aliphatic amines are gases with a fishy smell; primary amines with three or more carbons are liquids; higher ones are solids. Aniline and other arylamines are colourless when pure but turn coloured on standing, because air oxidises them.
- **Solubility.** Lower amines dissolve in water because they form **hydrogen bonds** with it. As the hydrophobic (water-repelling) alkyl part grows, solubility drops, so higher amines are insoluble. Amines dissolve in organic solvents (alcohol, ether, benzene). An alcohol is **more** soluble in water than the amine of similar size, because O is more electronegative than N, so the O-H bond is more polar and bonds more strongly to water.

### Boiling point

Primary and secondary amines hydrogen-bond to *each other* (N of one molecule to H of the next). A

**primary** amine has two N-H bonds available, a **secondary** has one, a **tertiary** has none. So, for isomeric amines:



And amines boil **lower than alcohols** of similar mass, because the O-H bond is more polar than N-H.

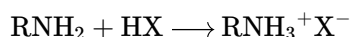
NCERT's numbers (all about 72–74 g/mol): *n*-butanol 390.3 K; *n*-butylamine 350.8 K; diethylamine 329.3 K; ethyldimethylamine 310.5 K; and the hydrocarbon isopentane 300.8 K (no hydrogen bonding at all).

**Trap:** the boiling-point and solubility questions are decided by the **number of N–H bonds and their polarity**, not by the molar mass. Primary amine, 2 N–H → highest. Tertiary amine, no N–H → lowest. Alcohol O–H beats any N–H. Aniline is the least soluble in water because of its large hydrophobic ring.

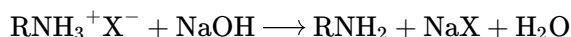
**Examiner asks:** arrange in increasing boiling point or solubility (2020 Q18; 2025 Q1, Q4); amine versus alcohol — which boils higher, and why (2020 Q3).

## Basicity: the idea — *K<sub>b</sub>, pK<sub>b</sub>, and one rule — the easier the lone pair is handed over, the stronger the base*

**[exposure]** A **base** here is something that accepts a proton H<sup>+</sup>. The nitrogen of an amine has an **unshared lone pair**, which it can donate to H<sup>+</sup>, so amines are **Lewis bases** (electron-pair donors). They turn acids into salts:



and the salts give the amine back when treated with a base:



Amine salts dissolve in water but not in ether. That is how an amine is **separated from non-basic organic compounds**: turn it into a water-soluble salt, wash away the rest, then add base to get it back.

**[exposure]** *K<sub>b</sub>* and *pK<sub>b</sub>* put a number on strength. In water an amine does  $\text{RNH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{RNH}_3^+ + \text{OH}^-$ . Since water is a pure liquid its concentration is constant and folds into the constant, giving the **base dissociation constant**:

$$K_b = \frac{[\text{RNH}_3^+][\text{OH}^-]}{[\text{RNH}_2]}, \quad pK_b = -\log K_b$$

**Larger *K<sub>b</sub>* ⇔ smaller *pK<sub>b</sub>* ⇔ stronger base.** The pK<sub>b</sub> scale runs *opposite* to basic strength — that reversal is where most marks are lost.

| Amine (aqueous)         | pK <sub>b</sub> | Amine (aqueous)                       | pK <sub>b</sub> |
|-------------------------|-----------------|---------------------------------------|-----------------|
| methanamine             | 3.38            | ethanamine                            | 3.29            |
| N-methylmethanamine     | 3.27            | N-ethylethanamine                     | 3.00            |
| N,N-dimethylmethanamine | 4.22            | N,N-diethylethanamine                 | 3.25            |
| ammonia                 | 4.75            | phenylmethanamine                     | 4.70            |
| benzenamine (aniline)   | 9.38            | N-methylaniline / N,N-dimethylaniline | 9.30 / 8.92     |

## Basicity: alkylamines, ammonia and aniline — *three comparisons, each with a different reason*

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### Alkylamine versus ammonia — the alkylamine wins

**[exposure]** The **+I effect** (positive inductive effect) is an alkyl group's habit of pushing electron density towards the atom it is attached to. On nitrogen it does two things: it makes the lone pair **more available** to take a proton, and it **stabilises the substituted ammonium ion** that results by spreading the positive charge. Both push the same way, so **alkylamines are stronger bases than ammonia** (pK<sub>b</sub> of ammonia 4.75; aliphatic amines 3 to 4.22).

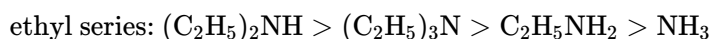
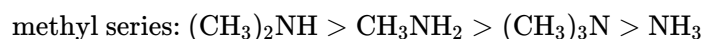
### Among alkylamines — gas phase versus water

With no solvent (gas phase) only the +I effect acts, so more alkyl groups means more basic: **tertiary > secondary > primary > NH<sub>3</sub>**.

In **water** the order is irregular, because three factors compete:

1. **+I effect** — favours tertiary > secondary > primary.
2. **[exposure] Solvation effect.** The protonated amine ion is stabilised when water molecules hydrogen-bond to its N–H hydrogens, releasing hydration energy. A primary ion (RNH<sub>3</sub><sup>+</sup>) has **three** N–H to bond with water, a secondary two, a tertiary one. So solvation favours primary > secondary > tertiary — the *reverse* of +I.
3. **[exposure] Steric hindrance.** Bulky groups crowd the nitrogen and block both the proton and the water. A small group like CH<sub>3</sub> blocks hydrogen bonding very little; an ethyl or larger group blocks it more. So changing methyl to ethyl changes the order.

The result, from NCERT:



In both series the **secondary amine is strongest**, because it has a good share of both +I and solvation. In the methyl series methyl is too small to cause steric hindrance, so solvation beats +I and primary > tertiary. In the ethyl series the bigger ethyls hinder solvation, so +I wins and tertiary > primary.

**Trap:** when a question gives no solvent, **assume aqueous**. And do not use one rule for both series — the methyl and ethyl orders differ (above). Memorise the two lines; they are the single most repeated ordering question (2019 Q11 gas phase; 2020 Q19(b); Q26(a)).

### Arylamine versus ammonia — aniline is weaker

In aniline the nitrogen is attached straight to the benzene ring, so its lone pair is **delocalised into the ring** by resonance (the +R effect). Delocalised electrons are **not available** for protonation. And once aniline *does* take a proton, the anilinium ion has only **two** resonance structures against **five** for aniline, so the ion is less stabilised. Result: aniline (pK<sub>b</sub> 9.38) is a much weaker base than ammonia (4.75).

## Substituents on aniline

**[exposure]** An **electron-releasing group (ERG)** such as  $-\text{CH}_3$  or  $-\text{OCH}_3$  raises the electron density on nitrogen and **increases** basic strength. An **electron-withdrawing group (EWG)** such as  $-\text{NO}_2$ ,  $-\text{CN}$ ,  $-\text{COOH}$  or a halogen pulls electron density away and **decreases** it.

**Examiner asks:** "arrange in order of basic strength" (NCERT intext 9.4, example 9.4; 2019 Q11; 2025 Q4(a)); "why is methylamine more basic than aniline / than ammonia" (2020 Q4, 2013 Q13(b)); "why  $(\text{CH}_3)_2\text{NH}$  is more basic than  $(\text{CH}_3)_3\text{N}$  in water" (2020 Q19(b); Q26(a)); "why amides are less basic than amines" — the nitrogen lone pair in an amide is delocalised onto the  $\text{C}=\text{O}$  oxygen (2025 Q1(b), Q16(b)).

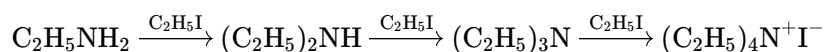
## Reactions of amines: salts, alkylation, acylation — *nitrogen as a nucleophile*

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An amine can act as a **nucleophile** (an electron-pair donor that attacks a positive centre) because of the lone pair; and the number of  $\text{N}-\text{H}$  bonds decides which reactions are possible.

### Alkylation

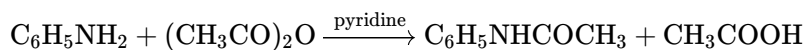
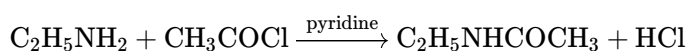
An amine reacts with an alkyl halide, replacing an  $\text{N}-\text{H}$  hydrogen with the alkyl group, so the amine moves up one class each time. It stops only when the nitrogen has four groups.



(N-ethylethanamine, N,N-diethylethanamine, then tetraethylammonium iodide.) Aniline does the same with  $\text{CH}_3\text{I}$ : N-methylaniline, N,N-dimethylaniline, and finally N,N,N-trimethylanilinium iodide.

### Acylation

**[exposure]** **Acylation** is replacing an  $\text{N}-\text{H}$  hydrogen of a **primary or secondary** amine by an **acyl group**  $\text{R}-\text{CO}-$ . The reagent is an acid chloride, an anhydride or an ester; the product is an **amide**. A **tertiary amine cannot** be acylated because it has no  $\text{N}-\text{H}$  to replace. Use a base **stronger than the amine**, such as **pyridine**: it mops up the  $\text{HCl}$  formed, shifting the equilibrium to the product side.



The second is aniline with ethanoic anhydride, giving **N-phenylethanamide, acetanilide**. With **benzoyl chloride**  $\text{C}_6\text{H}_5\text{COCl}$  the same reaction is called **benzoylation**:

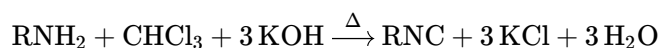
$\text{CH}_3\text{NH}_2 + \text{C}_6\text{H}_5\text{COCl} \longrightarrow \text{CH}_3\text{NHCOC}_6\text{H}_5 + \text{HCl}$  (N-methylbenzamide). Aniline plus benzoyl chloride gives N-phenylbenzamide.

**Trap:** the role of pyridine is a standing 1-mark question. Answer: it **neutralises the  $\text{HCl}$**  produced, which shifts the equilibrium forward so the reaction goes to completion.

## Carbylamine test, nitrous acid and Hinsberg — *three ways to tell the classes apart*

### Carbylamine reaction (the test for primary amines)

**[exposure]** An **isocyanide** (carbylamine) is  $R-N\equiv C$ , a nitrogen–carbon triple unit attached through nitrogen. Heating a **primary** amine (aliphatic or aromatic) with **chloroform and ethanolic KOH** gives an isocyanide with a **foul smell**. Secondary and tertiary amines do not give it, so it is the test for a primary amine.



Aniline gives phenyl isocyanide  $C_6H_5NC$  the same way.

### Reaction with nitrous acid

**[exposure]** **Nitrous acid**  $HNO_2$  is unstable, so it is made **in situ** ("in place", inside the flask) from sodium nitrite and dilute hydrochloric acid. The three classes behave differently:

- **Primary aliphatic** amine  $\rightarrow$  an aliphatic diazonium salt, which is **unstable** (no resonance) and breaks down to **nitrogen gas and an alcohol**:  $RNH_2 + HNO_2 \rightarrow ROH + N_2 + H_2O$ . The nitrogen released is measured to estimate amino acids and proteins.
- **Primary aromatic** amine at **273–278 K**  $\rightarrow$  a **diazonium salt**, which is stable at that temperature:  $C_6H_5NH_2 + NaNO_2 + 2 HCl \rightarrow C_6H_5N_2^+ Cl^- + NaCl + 2 H_2O$ .
- **Secondary and tertiary** amines react "in a different manner" (no nitrogen gas as with primary). The lecture stops at that statement.

The key exam use: **only primary aliphatic amines give off nitrogen gas** with nitrous acid — the NCERT  $C_3H_9N$  question (intext 9.8) turns on it: propan-1-amine and propan-2-amine give  $N_2$ ; the secondary and tertiary isomers do not.

### Hinsberg's reagent

**[exposure]** **Hinsberg's reagent** is **benzenesulphonyl chloride**,  $C_6H_5SO_2Cl$ . It distinguishes the three amine classes and also separates their mixtures. (Nowadays *p*-toluenesulphonyl chloride replaces it.)

| Amine             | Reaction           | Product                                                      | In aqueous alkali           |
|-------------------|--------------------|--------------------------------------------------------------|-----------------------------|
| Primary $RNH_2$   | reacts, loses HCl  | $C_6H_5SO_2NHR$ (N-alkylbenzenesulphonamide) has an N–H left | <b>soluble</b> (acidic N–H) |
| Secondary $R_2NH$ | reacts, loses HCl  | $C_6H_5SO_2NR_2$ (N,N-dialkylbenzenesulphonamide), no N–H    | <b>insoluble</b>            |
| Tertiary $R_3N$   | <b>no reaction</b> | —                                                            | —                           |

Why the primary sulphonamide dissolves: the strongly electron-withdrawing  $-SO_2-$  group makes the remaining N–H hydrogen **acidic**, so alkali removes it and the compound becomes a water-soluble salt. The secondary product has no N–H, so it cannot.

**Trap:** the answer for "distinguish primary, secondary, tertiary" needs **both** halves — reaction or not, *and* solubility of the product in alkali. For primary versus secondary alone, the **carbylamine test** is the quicker answer (2020 Q14; but N-methylaniline, being secondary, does not give it, while aniline does).

**Examiner asks:** the carbylamine equation (2025 Q2(b), 2022 Q28, 2014 Q25); distinguish primary from secondary (2020 Q14); distinguish secondary from tertiary with Hinsberg (2013 Q30(a)); "what is Hinsberg's reagent" (Q26(b)).

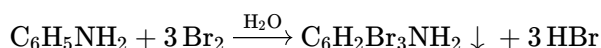
## Ring reactions of aniline — $\text{—NH}_2$ *is a powerful activator; that is both a gift and a nuisance*

---

**[exposure] Electrophilic substitution** is the reaction where an electron-seeking group (an **electrophile**) replaces a hydrogen on a benzene ring. A group already on the ring either **activates** it (more reactive than benzene) or **deactivates** it, and it steers the incoming group to a position: **ortho** (next to it), **para** (opposite), or **meta**. In aniline,  $\text{—NH}_2$  donates its lone pair into the ring, so the ortho and para positions become rich in electrons.  $\text{—NH}_2$  is a **powerful activating, ortho/para-directing group**, so aniline reacts faster than benzene (2013 Q16(a)).

### Bromination — too much of a good thing

Aniline reacts with bromine water at room temperature to give a **white precipitate of 2,4,6-tribromoaniline** (three bromines at once).



### Controlling the activation — protect the amino group

To get just one substituent, **acetylate** the  $\text{—NH}_2$  first (acetic anhydride, pyridine) to acetanilide  $\text{C}_6\text{H}_5\text{NHCOCH}_3$ . The nitrogen's lone pair is now shared with the  $\text{C=O}$  oxygen by resonance, so it is *less* available to the ring, and the activation is weaker. Carry out the substitution, then **hydrolyse** the amide back to the amine.

Worked example · aniline to 4-bromoaniline

Acetylate:  $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{(\text{CH}_3\text{CO})_2\text{O}} \text{C}_6\text{H}_5\text{NHCOCH}_3$ . Brominate in ethanoic acid: the bromine goes mostly **para** (the symmetric position, the major product). Hydrolyse with acid or base:  $\text{—NHCOCH}_3 \longrightarrow \text{—NH}_2$ . Result: **4-bromoaniline**.

### Nitration — two problems

Direct nitration with  $\text{HNO}_3 + \text{H}_2\text{SO}_4$  goes wrong in two ways. Nitric acid is a strong oxidant, so aniline is oxidised to **tarry products**. And in the strongly acidic mixture aniline picks up  $\text{H}^+$  to become the **anilinium ion**  $\text{C}_6\text{H}_5\text{NH}_3^+$ , which is **meta-directing** (the positive nitrogen withdraws electrons and deactivates the ortho/para positions). NCERT's yield at 288 K: **47% meta**, 51% para, 2% ortho. So aniline gives meta-nitroaniline in good yield even though  $\text{—NH}_2$  is itself o,p-directing (2020 Q19(a)).

The fix is the same: **acetylate first**, nitrate (para-nitroacetanilide is the main product), then hydrolyse to **p-nitroaniline**.

### Sulphonation

Aniline with concentrated  $\text{H}_2\text{SO}_4$  gives anilinium hydrogensulphate,  $\text{C}_6\text{H}_5\text{NH}_3^+\text{HSO}_4^-$ . Heated at **453–473 K** it rearranges to **p-aminobenzenesulphonic acid, sulphanilic acid**. In water it exists as a **zwitterion**,  $^-\text{O}_3\text{S—C}_6\text{H}_4\text{—NH}_3^+$  (acidic  $\text{—SO}_3\text{H}$  gives up  $\text{H}^+$ ; basic  $\text{—NH}_2$  takes it): 2023 Q32(a).

## No Friedel–Crafts

**[exposure] Friedel–Crafts** reactions (alkylation and acylation of a benzene ring) use  $\text{AlCl}_3$ , a Lewis acid, as the catalyst. Aniline does **not** undergo them: its nitrogen lone pair attacks the electron-deficient  $\text{AlCl}_3$  and forms a **salt**. The nitrogen is now positively charged, so it strongly withdraws electrons and **deactivates** the ring; no electrophile can attack. This is a 1- or 2-mark question that recurs (2014 Q13, 2017, 2020 Q29(a)).

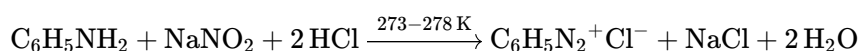
## Diazonium salts: what they are and how they are made — $\text{ArN}_2^+\text{X}^-$ — *one of the most useful functional groups in the course*

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**[exposure]** A **diazonium salt** has the general formula  $\text{ArN}_2^+\text{X}^-$ : an aryl group bonded to the  $-\text{N}_2^+$  **diazonium group**, with an anion  $\text{X}^-$  ( $\text{Cl}^-$ ,  $\text{Br}^-$ ,  $\text{HSO}_4^-$ ,  $\text{BF}_4^-$ ). They exist because  $\text{N}_2$  is such a good **leaving group** (a group that departs easily, here as nitrogen gas) that the ring can then be given almost any other group. Name them as the parent hydrocarbon + **diazonium** + the anion:  $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$  is **benzenediazonium chloride**;  $\text{C}_6\text{H}_5\text{N}_2^+\text{HSO}_4^-$  is benzenediazonium hydrogensulphate;  $\text{C}_6\text{H}_5\text{N}_2^+\text{BF}_4^-$  is benzenediazonium fluoroborate.

### Preparation — diazotisation

Dissolve aniline in  $\text{HCl}$  and add  $\text{NaNO}_2$  at **273–278 K** (0–5 °C). Nitrous acid forms in the flask and converts the primary aromatic amine into the diazonium salt. This is called **diazotisation**.



The salt is not stored; it is **used straight away**.

### Stability

An **aromatic** diazonium salt is far more stable than an aliphatic one. In the arenediazonium ion the positive charge is **spread over the benzene ring by resonance**; in an alkyl diazonium ion nothing can spread it, so it loses  $\text{N}_2$  at once to give a carbocation. Even so the aromatic one is stable only for a short time, in solution, at **273–278 K** (2018 Q17(b)).

### Physical properties

Benzenediazonium chloride is a colourless crystalline solid, soluble in water, stable in the cold, and it reacts with warm water; dry it decomposes easily. The **fluoroborate is insoluble in water and stable at room temperature**, which is why it is used for fluorine and nitro replacements.

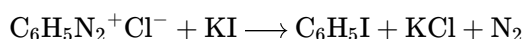
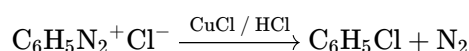
## Diazonium reactions: displacing nitrogen — *swap $\text{N}_2$ for Cl, Br, CN, I, F, H, OH or NO<sub>2</sub> — one reagent per swap*

---

The group  $-\text{N}_2^+$  leaves as nitrogen gas, and the group you want to put on takes its place. The table is the whole lesson:

| You want         | Reagent                                                                | Name                             |
|------------------|------------------------------------------------------------------------|----------------------------------|
| –Cl              | CuCl / HCl (or Cu / HCl)                                               | <b>Sandmeyer</b> (or Gattermann) |
| –Br              | CuBr / HBr (or Cu / HBr)                                               | <b>Sandmeyer</b> (or Gattermann) |
| –CN              | CuCN / KCN                                                             | <b>Sandmeyer</b>                 |
| –I               | KI                                                                     | (no copper needed)               |
| –F               | HB <sub>4</sub> F then heat                                            | via the fluoroborate             |
| –H               | H <sub>3</sub> PO <sub>2</sub> (or CH <sub>3</sub> CH <sub>2</sub> OH) | reduction                        |
| –OH              | warm water (up to 283 K)                                               | hydrolysis → phenol              |
| –NO <sub>2</sub> | HB <sub>4</sub> F, then NaNO <sub>2</sub> / Cu, heat                   | via the fluoroborate             |

**[exposure]** The **Sandmeyer reaction** introduces Cl<sup>–</sup>, Br<sup>–</sup> or CN<sup>–</sup> using a **copper(I)** salt. The **Gattermann reaction** does the chlorine/bromine version with **copper powder and the halogen acid** instead. The Sandmeyer gives the better yield (NCERT). Both are named reactions you can be asked by name.



**Fluorine:**  $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{HB}_4\text{F} \longrightarrow \text{C}_6\text{H}_5\text{N}_2^+\text{BF}_4^- + \text{HCl}$  ; the fluoroborate precipitates and, when heated, gives  $\text{C}_6\text{H}_5\text{F} + \text{BF}_3 + \text{N}_2$  .

**Hydrogen:**  $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{H}_3\text{PO}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_6 + \text{N}_2 + \text{H}_3\text{PO}_3 + \text{HCl}$  . The H<sub>3</sub>PO<sub>2</sub> is the reducing agent and is itself oxidised to H<sub>3</sub>PO<sub>3</sub> . Ethanol does the same job and is oxidised to ethanal CH<sub>3</sub>CHO .

**Hydroxyl:** warm the diazonium solution above the preparation temperature (up to 283 K) and it is hydrolysed:  $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_5\text{OH} + \text{N}_2 + \text{HCl}$  . This is why diazotisation is kept at 273–278 K – a little warmer and phenol forms (2013 Q1: "benzenediazonium chloride + warm water").

**Nitro:**  $\text{C}_6\text{H}_5\text{N}_2^+\text{BF}_4^- + \text{NaNO}_2 \xrightarrow{\text{Cu}, \Delta} \text{C}_6\text{H}_5\text{NO}_2 + \text{N}_2 + \text{NaBF}_4$  .

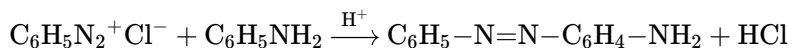
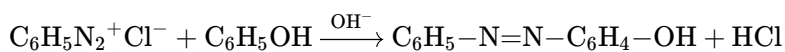
**Trap:** I and F are the awkward ones. Iodine and fluorine cannot be put on the ring by direct halogenation (fluorine is too violent, iodine too unreactive), and –CN cannot be put on by substituting chlorobenzene.

**Diazonium salts are the way to make aryl fluorides, aryl iodides and aryl cyanides.** Say so when asked why the diazonium route is used (Q on the importance of diazonium salts).

## Diazonium reactions: coupling — *the –N=N– link keeps its nitrogen — and makes dyes*

**[exposure]** A **coupling reaction** joins the diazonium ion to another electron-rich benzene ring *without losing the nitrogen*. The positive diazonium end is the electrophile; the partner (phenol or aniline) has a ring made electron-rich by its –OH or –NH<sub>2</sub>, and the ring is attacked at the **para** position. The product is an

**azo compound**, two aromatic rings joined by  $-\text{N}=\text{N}-$ . That link carries an extended system of alternating double bonds (a **conjugated system**), which absorbs visible light, so these compounds are **coloured and used as dyes**.



- With **phenol** (in a basic solution, where the phenol is present as the more reactive phenoxide): **p-hydroxyazobenzene**, an **orange dye**.
- With **aniline** (acidic medium): **p-aminoazobenzene**, a **yellow dye**.

Coupling is itself an **electrophilic substitution** on the second ring.

**Examiner asks:** the coupling of benzenediazonium chloride with aniline (2025 Q3(b)); the displacement reactions (2019 Q24; 2020 Q10; 2014 Q15; 2013 Q1); "diazotisation with an equation" (2018 Q17(a)); "why aromatic diazonium salts are more stable than aliphatic" (2018 Q17(b)).

## Diazonium conversions — *three NCERT multi-step problems, each built from the table above*

---

Worked example · NCERT Example 9.5 — 4-nitrotoluene to 2-bromobenzoic acid

1. Brominate: the  $\text{CH}_3$  is ortho/para-directing and the para place is taken by  $\text{NO}_2$ , so Br goes **ortho to the methyl**.
2. Remove the  $\text{NO}_2$ : reduce it to  $\text{NH}_2$  ( $\text{Sn}/\text{HCl}$ ), diazotise ( $\text{NaNO}_2/\text{HCl}$ , 273–278 K), and replace the diazonium group by hydrogen ( $\text{H}_3\text{PO}_2$ ).
3. Oxidise the  $\text{CH}_3$  to  $\text{COOH}$  ( $\text{KMnO}_4$ , base).

The diazonium route is how an  $\text{NH}_2$  (and so an  $\text{NO}_2$ ) is removed.

Worked example · NCERT intext 9.9 — 3-methylaniline to 3-nitrotoluene

Diazotise ( $\text{NaNO}_2 / \text{HCl}$ , 273–278 K); make the fluoroborate ( $\text{HBF}_4$ ); then  $\text{NaNO}_2 / \text{Cu}$ , heat — the diazonium group is replaced by  $\text{NO}_2$ .

Worked example · NCERT intext 9.9 — aniline to 1,3,5-tribromobenzene

Bromine water first:  $-\text{NH}_2$  directs three Br to **2,4,6**, giving 2,4,6-tribromoaniline. Then diazotise and replace the diazonium group by hydrogen ( $\text{H}_3\text{PO}_2$ ). With  $\text{NH}_2$  gone, the three bromines are at 1,3,5.

## Patterns, collected — *the orderings, tests and conversions you will actually be asked to produce*

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### 1. Rank by basic strength

Worked example

Arrange in decreasing basic strength:  $\text{C}_6\text{H}_5\text{NH}_2$ ,  $\text{C}_2\text{H}_5\text{NH}_2$ ,  $(\text{C}_2\text{H}_5)_2\text{NH}$ ,  $\text{NH}_3$  (NCERT Example 9.4).

1. No solvent named, so use the aqueous order. The ethyl series:  $(\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3$ .

2. Aniline's lone pair is delocalised into the ring, so aniline goes last.

Answer:  $(\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3 > \text{C}_6\text{H}_5\text{NH}_2$ . For  $\text{pK}_\text{b}$ , reverse the order.

- Increasing basic strength:  $\text{CH}_3\text{NH}_2$ ,  $(\text{CH}_3)_3\text{N}$ ,  $(\text{CH}_3)_2\text{NH}$  in water — answer only:  
 $(\text{CH}_3)_3\text{N} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$
- Gas-phase order of primary, secondary, tertiary — answer only: tertiary > secondary > primary  
(increasing:  $1^\circ < 2^\circ < 3^\circ$ )
- Increasing  $\text{pK}_\text{b}$ :  $(\text{C}_2\text{H}_5)_2\text{NH}$ ,  $\text{NH}_3$ ,  $\text{C}_2\text{H}_5\text{NH}_2$ ,  $\text{C}_6\text{H}_5\text{NH}_2$  — answer only:  
 $(\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2 < \text{NH}_3 < \text{C}_6\text{H}_5\text{NH}_2$

## 2. Rank by boiling point or solubility

Worked example

Increasing boiling point: a secondary amine, a primary amine and ethanol (2025 Q1(a)).

Count and compare the hydrogen bonds: secondary has one N–H, primary has two, ethanol has an O–H that is more polar than any N–H. Order: secondary < primary < ethanol.

- Increasing solubility in water: aniline, a secondary amine, a primary amine — answer only: aniline < secondary < primary
- Higher boiling point: propan-1-amine or N,N-dimethylmethanamine — answer only: propan-1-amine (primary, hydrogen bonds)

## 3. Distinguish by a test

Worked example

Distinguish ethylamine from diethylamine (2020 Q14).

Heat each with chloroform and ethanolic KOH. Ethylamine (primary) gives the foul-smelling ethyl isocyanide; diethylamine (secondary) does not.

- Distinguish a secondary from a tertiary amine — answer only: Hinsberg's reagent; the secondary gives an alkali-insoluble sulphonamide, the tertiary does not react
- Distinguish aniline from N-methylaniline — answer only: carbylamine test; aniline (primary) gives it, N-methylaniline (secondary) does not

## 4. Convert, tracking the carbon count

Worked example

Ethanamide to methanamine (2022 Q21): one carbon fewer, so Hofmann:  $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{Br}_2 / \text{NaOH}} \text{CH}_3\text{NH}_2$ .

- Bromoethane to propan-1-amine — answer only: KCN to give  $\text{CH}_3\text{CH}_2\text{CN}$ , then  $\text{LiAlH}_4$  (or  $\text{H}_2/\text{Ni}$ ) to give  $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- Benzene to aniline — answer only: nitration ( $\text{HNO}_3/\text{H}_2\text{SO}_4$ ), then  $\text{Fe}/\text{HCl}$

## 5. Pick the diazonium reagent

Worked example

Aniline to chlorobenzene (2025 Q5(b)): diazotise with  $\text{NaNO}_2 / \text{HCl}$  at 273–278 K to  $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ , then  $\text{CuCl} / \text{HCl}$  (Sandmeyer) to  $\text{C}_6\text{H}_5\text{Cl}$ .

- Benzenediazonium chloride to iodobenzene — answer only: KI

- Benzenediazonium chloride to phenol — answer only: warm water
- Benzenediazonium chloride to benzene — answer only:  $\text{H}_3\text{PO}_2$  (or ethanol)

## 6. Count the isomers, then test them with nitrous acid

Worked example

$\text{C}_3\text{H}_9\text{N}$ : write the isomers and say which release nitrogen with nitrous acid (NCERT intext 9.8).

Primary: propan-1-amine, propan-2-amine. Secondary: N-methylethanamine. Tertiary: N,N-dimethylmethanamine. Only the primary aliphatic amines give nitrogen gas: propan-1-amine and propan-2-amine.

## Past year questions · question types — ranked by how often they turn up

Two videos feed this section. [30+] is the "30+ PYQ" video for CBSE 2024-25 — Q1–Q33, questions from 2013 to 2023. [2025] is the short "2025 PYQ" video — Q1–Q5, from the 2025 board paper. Question numbers are each video's own numbering, and years come from the video. Where the video gave no year, the question is cited by number only. The CBSE 2026-27 "25 PYQ" video for this chapter was not uploaded when these notes were built; once it is, this section can be refreshed from it. Counts are by question part, from the two videos' lists.

1 · Convert, complete, or identify A, B, C — 1 to 3 marks · about 22 parts

Recognise it: "how will you convert ...", "complete the reaction", "identify A, B and C", "give the equation for ...", "write the structure of the product".

1. Name the type of change: nitro  $\rightarrow$  amine; nitrile  $\rightarrow$  amine; amide  $\rightarrow$  amine; amine  $\rightarrow$  diazonium salt.
2. **Count carbons** in the start and the product: more carbons  $\rightarrow$  nitrile route; fewer  $\rightarrow$  Hofmann.
3. Write each step with its reagent over the arrow; for A/B/C questions, write the formula and name of each letter.

**Trap:** the reagents decide the marks. Nitro  $\rightarrow$  amine is  $\text{Fe}/\text{HCl}$  or  $\text{Sn}/\text{HCl}$  (not  $\text{LiAlH}_4$ ); nitrile  $\rightarrow$  amine is  $\text{LiAlH}_4$  or  $\text{H}_2/\text{Ni}$ ; Hofmann needs  $\text{Br}_2 + \text{NaOH}$ ; diazotisation is  $\text{NaNO}_2 + \text{HCl}$  at **0–5 °C** (273–278 K) — at room temperature you get phenol.

2 · Give a reason / account for — 1 to 3 marks · about 14 parts

Recognise it: "give reason ...", "account for ...", "why is X more basic / weaker / unable to ...".

1. State the structural fact: a lone pair on N; resonance; N–H count; an electron-pulling or -pushing group.
2. Say how it causes the behaviour in one sentence.

**Trap:** the same three answers cover most of these: **aniline's lone pair is delocalised into the ring** (weak base; no Friedel–Crafts; ring is activated); **aryl halides do not do nucleophilic substitution** (no Gabriel for aromatic amines); and **the solvation effect** (secondary > tertiary in water).

3 · Arrange in order — 1 to 3 marks · about 10 parts

Recognise it: "arrange in increasing/decreasing order of basic strength /  $\text{pK}_\text{b}$  / boiling point / solubility in water", "in the gaseous phase".

1. Decide which property it is and what controls it: basicity → lone-pair availability; boiling point and solubility → hydrogen bonding.
2. Basicity: no solvent named → aqueous. Use the two NCERT orders for methyl and ethyl series; aniline is last;  $pK_b$  runs the opposite way.
3. Boiling point: primary > secondary > tertiary, and alcohol above all of them. Solubility: more N–H and a smaller ring/chain → more soluble.

**Trap:** read "increasing" or "decreasing", and read  $pK_b$  versus basic strength — a smaller  $pK_b$  means a *stronger* base. Gas phase is the only case where tertiary > secondary > primary.

4 · Distinguish between two compounds — *2 or 3 marks · about 5 parts*

*Recognise it: "distinguish between ..." or "what is Hinsberg's reagent".*

1. Primary versus secondary or tertiary: carbylamine test (chloroform + ethanolic KOH, foul smell, primary only).
2. Secondary versus tertiary: Hinsberg's reagent  $C_6H_5SO_2Cl$  — secondary gives a sulphonamide that is insoluble in alkali; tertiary does not react.

**Trap:** aniline is a primary amine and does give the carbylamine test; N-methylaniline is secondary and does not. Do not treat "aromatic" as "different".

5 · Write the IUPAC name — *1 or 2 marks · 4 parts*

*Recognise it: a drawn amine, with "write the IUPAC name".*

1. Find the longest carbon chain on nitrogen; that is the parent (...anamine). For a ring on N it is aniline.
2. Every other group on nitrogen becomes an N-substituent, alphabetical order.

**Trap:** in 2,4,6-tribromoaniline the  $-NH_2$  carbon is number 1; do not number from a bromine.

## Past year questions · mark slots — *what each type is worth*

The 30+ video states its own marks: Q1–Q12 are 1-mark, Q13–Q20 are 2-mark, Q21–Q30 are 3-mark, and the three 2023 questions Q31–Q33 are stated as 2-mark each. The 2025 video states Q1, Q2 and Q4 as 3-mark and Q3 as 2-mark, and does not state the marks for Q5 (inferred: 3). Amines does not appear as a 5-mark long answer in either video.

| Question type                 | 1 mark                                                  | 2 marks                                                      | 3 marks                                                                    |
|-------------------------------|---------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------|
| Convert / complete / identify | 2019 Q9, 2020 Q10, 2014 Q12, 2013 Q1, 2015 Q7, Q5 [30+] | 2014 Q15, 2018 Q20, 2023 Q31(a)(b) [30+]; 2025 Q3 [2025]     | 2022 Q21, 2015 Q23, 2019 Q24, 2014 Q25, 2022 Q28 [30+]; 2025 Q2, Q5 [2025] |
| Give a reason                 | 2020 Q3, Q4, Q8 [30+]                                   | 2014 Q13, 2013 Q16, 2018 Q17(b), 2020 Q19, 2023 Q32(b) [30+] | Q29, 2013 Q30(b) [30+]; 2025 Q1(b) [2025]                                  |
| Arrange in order              | 2019 Q11 [30+]                                          | 2020 Q18 [30+]                                               | Q26(a), Q27(b) [30+]; 2025 Q1(a), Q4 [2025]                                |
| Distinguish / reagent         | —                                                       | 2020 Q14 [30+]                                               | 2013 Q30(a), Q26(b) [30+]                                                  |
| IUPAC name                    | 2017 Q2, 2016 Q6 [30+]                                  | 2023 Q33(a) [30+]                                            | 2022 Q22 [30+]                                                             |

## Past year questions · repeat offenders — *appeared more than once* — *highest probability in the chapter*

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2019 Q9 · 2022 Q21(b) · 2015 Q23 · 2014 Q15 · 2014 Q25

Hofmann bromamide degradation — the equation, ethanamide to methanamine, benzamide to aniline. The single most repeated reaction.

2014 Q13(a) · 2017 · 2020 Q29(a)

Why aniline does not undergo Friedel–Crafts: it forms a salt with  $\text{AlCl}_3$ , the nitrogen becomes positive and deactivates the ring.

2017 · 2020 Q29(b) · 2025 Q1(b)

Why aromatic primary amines cannot be made by Gabriel's synthesis: aryl halides do not undergo nucleophilic substitution (partial double-bond character).

2020 Q14 · 2022 Q28 · 2014 Q25 · 2025 Q2(b)

Carbylamine test / phenyl isocyanide from aniline — foul smell, primary amines only.

2015 Q7 · 2022 Q21(c) · 2023 Q31(b) · 2014 Q25 · 2025 Q3(a)

Nitrile reduction to an amine, often as the second step after  $\text{KCN}/\text{NaCN}$  — the "go up one carbon" route.

2013 Q1 · 2014 Q15 · 2025 Q5(c)

Benzenediazonium chloride warmed with water gives phenol (and why diazotisation must stay at 0–5 °C).

2020 Q10 · 2019 Q24 · 2025 Q5(b)

Diazonium chloride with  $\text{CuCN}$  (benzonitrile → benzoic acid on hydrolysis),  $\text{KI}$  (iodobenzene),  $\text{CuCl}/\text{HCl}$  (chlorobenzene).

2019 Q24 · 2023 Q31(a)

Diazonium chloride with ethanol (or hypophosphorous acid) gives benzene; the ethanol is oxidised to ethanal.

2020 Q5 · 2014 Q15 · 2022 Q21(a) · 2022 Q28 · 2023 Q33(b)

Nitrobenzene reduced to aniline with  $\text{Fe}/\text{HCl}$  or  $\text{Sn}/\text{HCl}$  — the first step in a long list of sequences.

2014 Q12 · 2018 Q20(a) · 2025 Q2(c) · 2023 Q32(b) · 2025 Q5(a)

Acetylation of aniline with acetic anhydride and pyridine (acetanilide), and its use to protect the  $\text{—NH}_2$  in nitration and bromination.

2020 Q19(b) · Q26(a) · 2019 Q11 · Q27(b) · 2025 Q4(a)

Basic-strength and  $\text{pK}_\text{b}$  orderings: secondary > tertiary in water (solvation + +I), tertiary highest in the gas phase, aniline last.

2020 Q3 · Q8 · 2020 Q18(b) · 2025 Q1(a) · 2025 Q4(b)

Boiling-point order: primary > secondary > tertiary, and an alcohol above an amine.

2013 Q16(b) · 2025 Q1(b)

Why amides are weaker bases than amines: the nitrogen lone pair is delocalised onto the carbonyl oxygen by resonance.

2025 Q2(a) · 2023 Q32(a)

Sulphanilic acid — how to make it from aniline, and its zwitterion.

2013 Q30(a) · Q26(b) · 2018 Q20(b)

Hinsberg's reagent — secondary versus tertiary; the sulphonamide of a secondary amine.

2017 Q2 · 2016 Q6 · 2022 Q22 · 2023 Q33(a)

An IUPAC name for a tertiary amine, a tribromoaniline, or a mixed N-substituted aniline.

## Past year questions · numerical types — *no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice*

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### A · Identify A, B, C in a sequence

Worked example · 2015 Q23 [30+] — A is  $\text{C}_7\text{H}_6\text{O}_2$ ; A + aqueous  $\text{NH}_3$  (heat) gives B; B with  $\text{Br}_2$  and alcoholic KOH gives C,  $\text{C}_6\text{H}_7\text{N}$

1.  $\text{Br}_2 + \text{KOH}$  is the Hofmann reagent, so B is an **amide** and C is the **amine** with one carbon fewer.  
 $\text{C}_6\text{H}_7\text{N}$  = aniline = **C**.

2. An aromatic amide with seven carbons is benzamide  $\text{C}_6\text{H}_5\text{CONH}_2$  = **B**.

3. An acid +  $\text{NH}_3$ , heat, gives the amide, so A is benzoic acid  $\text{C}_6\text{H}_5\text{COOH}$  ( $\text{C}_7\text{H}_6\text{O}_2$  ✓).



- 2014 Q15 [30+] — nitrobenzene with  $\text{Sn}/\text{HCl}$ , then  $\text{NaNO}_2/\text{HCl}$  at 273 K, then warm water — answer only: A aniline; B benzenediazonium chloride; C phenol
- 2014 Q25 [30+] —  $\text{CH}_3\text{CH}_2\text{Br}$  with KCN, then  $\text{LiAlH}_4$ , then  $\text{HNO}_2$  at 0 °C — answer only:  $\text{CH}_3\text{CH}_2\text{CN}$ ;  $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ ; propan-1-ol
- 2022 Q28 [30+] — A with iron scrap and HCl gives  $\text{C}_6\text{H}_7\text{N}$ ; with  $\text{CHCl}_3$  and alcoholic KOH gives a foul smell — answer only: A nitrobenzene; B aniline; C phenyl isocyanide
- 2025 Q3(a) [2025] — chloroethane + NaCN to A, then  $\text{H}_2/\text{Ni}$  to B — answer only: A  $\text{CH}_3\text{CH}_2\text{CN}$ ; B propan-1-amine

## B • Replace the diazonium group

Worked example • 2019 Q24 [30+] — benzenediazonium chloride with CuCN, with ethanol, and with KI

CuCN: CN takes the place of N<sub>2</sub> → **benzonitrile** C<sub>6</sub>H<sub>5</sub>CN (Sandmeyer). **Ethanol** is a mild reducing agent: hydrogen takes the place of N<sub>2</sub> → **benzene**; the ethanol is oxidised to ethanal. KI: iodide takes the place of N<sub>2</sub> → **iodobenzene**. Each time nitrogen gas leaves.

- 2020 Q10 [30+] — benzenediazonium chloride + CuCN gives A; A hydrolysed gives B — answer only: A benzonitrile; B benzoic acid
- 2013 Q1 [30+] — benzenediazonium chloride warmed with water — answer only: phenol (+ N<sub>2</sub> + HCl)
- 2023 Q31(a) [30+] — benzenediazonium chloride with ethanol — answer only: benzene; ethanol is oxidised to ethanal
- 2025 Q5(b) [2025] — chlorobenzene from aniline — answer only: NaNO<sub>2</sub>/HCl at 0–5 °C, then CuCl/HCl

## C • Substitute on aniline's ring (protect, react, deprotect)

Worked example • 2025 Q5(a) [2025] — *p*-nitroaniline from aniline

Acetylate: C<sub>6</sub>H<sub>5</sub>NH<sub>2</sub>  $\xrightarrow{(\text{CH}_3\text{CO})_2\text{O}}$  C<sub>6</sub>H<sub>5</sub>NHCOCH<sub>3</sub>. Nitrate with HNO<sub>3</sub>/H<sub>2</sub>SO<sub>4</sub>: the para product is the major one, *p*-nitroacetanilide. Hydrolyse with acid or base to get back –NH<sub>2</sub>: ***p*-nitroaniline**. The acetyl group stops oxidation and stops the anilinium ion forming, so the nitro group goes para, not meta.

- 2025 Q2(a) [2025] — sulphanilic acid from aniline — answer only: aniline + conc. H<sub>2</sub>SO<sub>4</sub> → anilinium hydrogensulphate; heat at 453–473 K
- 2025 Q2(c) / 2014 Q12 [2025, 30+] — acetanilide from aniline — answer only: (CH<sub>3</sub>CO)<sub>2</sub>O with pyridine → N-phenylethanamide
- 2023 Q33(b) [30+] — nitrobenzene to 2,4,6-tribromoaniline — answer only: Sn/HCl to aniline, then Br<sub>2</sub> water

## D • Arrange in order

Worked example • 2020 Q18 [30+] — increasing solubility in water: aniline, a secondary amine, a primary amine

Solubility rests on hydrogen bonding with water. The primary amine has two N–H, the secondary one, and aniline has a large hydrophobic ring. Order: **aniline < secondary < primary**. (Boiling point, same question: tertiary < secondary < primary.)

- 2019 Q11 [30+] — basic strength in the gas phase, increasing — answer only: primary < secondary < tertiary
- Q26(a) [30+] — CH<sub>3</sub>NH<sub>2</sub>, (CH<sub>3</sub>)<sub>3</sub>N, (CH<sub>3</sub>)<sub>2</sub>NH in water, increasing — answer only: (CH<sub>3</sub>)<sub>3</sub>N < CH<sub>3</sub>NH<sub>2</sub> < (CH<sub>3</sub>)<sub>2</sub>NH
- Q27(b) [30+] — increasing pK<sub>b</sub>: (C<sub>2</sub>H<sub>5</sub>)<sub>2</sub>NH, C<sub>2</sub>H<sub>5</sub>NH<sub>2</sub>, NH<sub>3</sub>, C<sub>6</sub>H<sub>5</sub>NH<sub>2</sub> — answer only: (C<sub>2</sub>H<sub>5</sub>)<sub>2</sub>NH < C<sub>2</sub>H<sub>5</sub>NH<sub>2</sub> < NH<sub>3</sub> < C<sub>6</sub>H<sub>5</sub>NH<sub>2</sub>
- 2025 Q4(a) [2025] — decreasing pK<sub>b</sub> of four amines — answer only: aniline first (highest pK<sub>b</sub>), then the methyl-substituted arylamine, then the primary, then the secondary aliphatic amine (the four structures are shown on screen, not spoken)

## E • Give a reason

Worked example • 2020 Q19(b) [30+] — (CH<sub>3</sub>)<sub>2</sub>NH is more basic than (CH<sub>3</sub>)<sub>3</sub>N in aqueous solution

Three factors decide basicity in water. **+I effect:** three methyls push more electron density onto nitrogen, favouring the tertiary. **Solvation:** the secondary cation has two N–H to hydrogen-bond with water, the tertiary only one, so the secondary cation is better stabilised. **Steric hindrance:** methyl is small, so it does not block the solvation. Solvation outweighs +I, so the **secondary amine is the stronger base**.

- 2014 Q13(a) / 2020 Q29(a) [30+] — aniline gives no Friedel–Crafts reaction — answer only: forms a salt with  $\text{AlCl}_3$ ; nitrogen turns positive and deactivates the ring
- 2018 Q17(b) [30+] — why aromatic diazonium salts are more stable than aliphatic — answer only: resonance spreads the positive charge over the benzene ring
- 2013 Q16(a) [30+] — electrophilic substitution in aromatic amines is easier than in benzene — answer only: the nitrogen lone pair is delocalised into the ring (+R), raising its electron density
- 2025 Q1(b)(ii) [2025] — amides are less basic than amines — answer only: the nitrogen lone pair is in resonance with the C=O group, so it is not available to take a proton

## F • Distinguish

Worked example • 2013 Q30(a) [30+] — distinguish a secondary from a tertiary amine

Add benzenesulphonyl chloride (Hinsberg's reagent). The **secondary** amine reacts and loses HCl, forming an N,N-dialkylbenzenesulphonamide that has no N–H, so it is **insoluble in alkali**. The **tertiary** amine has no hydrogen on nitrogen, so there is no reaction.

- 2020 Q14 [30+] — ethylamine versus diethylamine — answer only: carbylamine test; ethylamine gives the foul-smelling isocyanide, diethylamine does not
- 2020 Q14 [30+] — aniline versus N-methylaniline — answer only: carbylamine test; aniline (primary) gives it, N-methylaniline does not

## G • IUPAC names

Worked example • 2022 Q22 [30+] — name  $(\text{CH}_3\text{CH}_2)_2\text{N}-\text{CH}_3$

Longest chain on nitrogen: two carbons (either ethyl) → **ethanamine**. The other ethyl and the methyl are N-substituents, alphabetical: **N-ethyl-N-methylethanamine**.

- 2017 Q2 [30+] —  $(\text{CH}_3)_2\text{N}-\text{CH}_2\text{CH}_3$  — answer only: N,N-dimethylethanamine
- 2016 Q6 [30+] — aniline with three bromines at the 2, 4, 6 positions — answer only: 2,4,6-tribromoaniline
- 2023 Q33(a) [30+] — a tertiary amine with ethyl, methyl and phenyl on nitrogen — answer only: N-ethyl-N-methylaniline